

# Week 4 – BSM Performance Benchmark Documentation

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This document records the Black-Scholes-Merton (BSM) model performance baseline for JPM options pricing over the 2018–2024 period. All figures here serve as the reference point for any future model enhancements.

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## 1. Benchmark Setup

| Parameter              | Value                   |
|------------------------|-------------------------|
| Underlying asset       | JPM (JPMorgan Chase)    |
| Evaluation period      | 2018-01-01 – 2024-12-31 |
| Sampling frequency     | Monthly (month-start)   |
| Maturities             | [0.25, 0.5, 1.0] years  |
| Moneyness levels (K/S) | [0.9, 1.0, 1.1]         |
| Option types           | Call, Put               |
| Total observations     | 1,008                   |
| MC benchmark paths     | 10,000 (seed=42)        |
| Historical vol window  | 20 trading days         |
| Risk-free rate source  | FRED DGS10              |
| Dividend yield         | Trailing-twelve-month   |

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## 2. Headline Baseline Metrics

These are the official BSM baseline figures. Future models must beat these numbers to demonstrate improvement.

| Metric                   | Baseline value      |
|--------------------------|---------------------|
| <b>**Overall MAE**</b>   | <b>**0.112135**</b> |
| <b>**Overall RMSE**</b>  | <b>**0.174916**</b> |
| Max absolute error       | 2.106593            |
| Low-VIX MAE (VIX < 20.0) | 0.087432            |
| Mid-VIX MAE (20.0–30.0)  | 0.137720            |

| Metric                                               | Baseline value      |
|------------------------------------------------------|---------------------|
| <b>**High-VIX MAE (VIX <math>\geq</math> 30.0)**</b> | <b>**0.203184**</b> |
| Low-VIX RMSE                                         | 0.128842            |
| Mid-VIX RMSE                                         | 0.183733            |
| High-VIX RMSE                                        | 0.344156            |
| Sentiment–error Pearson corr                         | 0.215158            |

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### 3. Full Breakdown by Group

All sub-group MAE / RMSE values for complete traceability:

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group n MAE RMSE max_abs_err
overall 1008 0.112135 0.174916 2.106593
regime=low 630 0.087432 0.128842 0.624741
regime=medium 288 0.137720 0.183733 0.756761
regime=high 90 0.203184 0.344156 2.106593
maturity=0.25yr 336 0.080286 0.120687 0.580171
maturity=0.5yr 336 0.104617 0.155197 1.430548
maturity=1.0yr 336 0.151502 0.230511 2.106593
type=call 504 0.130680 0.209066 2.106593
type=put 504 0.093590 0.132221 0.583721
moneyness=0.9 336 0.108003 0.159285 0.756761
moneyness=1.0 336 0.112518 0.159001 0.624741
moneyness=1.1 336 0.115883 0.202814 2.106593
```

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### 4. Key Limitations Identified

1. **\*\*High-volatility failure\*\***: MAE in high-VIX regime (0.2032) is 232% of the low-VIX baseline (0.0874).  
BSM underprices risk during market stress because it uses backward-looking historical volatility rather than forward-looking implied volatility.
2. **\*\*Maturity effect\*\***: Error scales with maturity (T=1yr RMSE is materially larger than T=0.25yr), reflecting accumulated GBM path uncertainty that

BSM's closed form cannot fully capture when  $\sigma$  is volatile.

3. **Sentiment gap**: Pearson correlation of 0.215158 between news sentiment and pricing error confirms BSM's inability to respond to information events. Positive-sentiment days show higher mean error (0.12149) vs negative-sentiment days (0.072511), suggesting bullish news events drive larger deviations from the simulation benchmark.

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## 5. Improvement Targets for Future Models

| Target                | Current baseline | Goal                                  |
|-----------------------|------------------|---------------------------------------|
| Overall MAE           | 0.112135         | < 0.089708 (–20%)                     |
| High-VIX MAE          | 0.203184         | < 0.152388 (–25%)                     |
| Sentiment correlation | 0.215158         | $\approx 0$ (model absorbs sentiment) |